

Mamoon Rashid, PhD

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ACADEMIC QUALIFICATIONS

PhD (2010) Bioinformatics, IMTECH (Chandigarh) and Jawaharlal Nehru University, New Delhi, India

M.Sc. (2005) Biotechnology, Guru Jambheshwar University, Hisar, Haryana, India

B.Sc. (2003) Chemistry, Magadh University, Bihar, India

RESEARCH AND TRAINING INTERESTS

Research

- Advanced Bioinformatics analyses to understand human diseases/phenotype
- Single cell genomics (Assembly of single cell genome)
- Multi-omics (Genomics, transcriptomics, proteomics, metabolomics) data analysis
- Next-generation sequencing (Whole Genome, Whole Exome, Transcriptome, Epigenome)
- Application of Artificial Intelligence on Biological data to develop prediction models

Training

- A passion for transferring knowledge to students and peers in computational biology
- Invoking interest for career paths in genomics and bioinformatics
- Sharing/developing ideas with students for cutting edge research in bioinformatics

WORK EXPERIENCE

1. King Abdullah International Medical Research Center (KAIMRC), Riyadh, Saudi Arabia**Research Scientist, 2016-present**

- *Aging and nanotechnology: for a better understanding of nanoparticles interaction with aged cells (NRC23R/464/08) [Role: Co-PI], Status: Started, Hevolution Grant Aging Research.*
- *Defining epigenetic regulatory mechanisms of human heat stress response with and without heatstroke [Role: PI, Status: Undergoing, Smart Health Initiative Grant from KAUST-KAIMRC, KSA]*
- *Omics-guided identification of neoantigens for cancer vaccine from breast cancer patients – a pilot study [Role: PI, Status: Undergoing, Funded by KAIMRC, NRC21R/082/03]*
- *Identification of disease-causing mutations by WGS that are not detected by WES in families with hereditary disorders RC16/211/R2 [Role: Co-PI, Status: Completed]*

- *Characterization of KAIMRC1 cell to discover novel breast cancer causing pathways RC17/153/R [Role: Co-PI, Status: Completed]*
- *Generation of a library of naturally immortalized KAIMRC cancer cells RC18/173/R [Role: Co-PI, Status: Completed]*
- *Role of RORgammaT Transcription Factor in the Double Positive Thymocytes Development and Transformation RA16/018/A [Role: Co-PI, Status: Completed]*
- *Transcriptome analysis of the Saudi patients suffering of Hyper Immunoglobulin E Syndrome (HIES): A surge for biomarker and therapeutic target discovery RA17/006/A [Role: Co-PI, Status: Completed]*
- *Application of a system biology approach to uncover the mechanisms of heat-induced exhaustion and stroke RC17/026/R [Role: Co-PI, Status: Undergoing]*
- *Epidemiology, genetic factors, and molecular signature of heatstroke: an integrated translational approach RC18/373/R [Role: Co-PI, Status: Undergoing]*

2. King Abdullah University of Science and Technology, Saudi Arabia Postdoctoral Research, 2010-2015

- Single cell sequence analysis: Development of computational pipeline for decontamination of draft genomes of bacterial and archaeal single cell from RED SEA brine pools.
- Development of software tool for identification of pathogenic or foreign sequences in the heap of sequence data from next generation sequencing platform (tool named [READSCAN](#)).
- Assembly and annotation of microbial genomes.
- Collaborative research with “FANTOM5 consortium”, RIKEN, Tokyo

3. Jawaharlal Nehru University, New Delhi, India Doctoral Research, 2005- 2010

Dissertation title: *Development of Bioinformatics tools for analysis and prediction of therapeutic peptides/proteins*

- Developed a computational model using support vector machine for prediction of subcellular localization of Mycobacterial proteins ([TBPRED](#))
- Database of hormones and receptors ([HMRBASE](#))
- A computational prediction system for protein-protein interaction using primary structure ([ProPrint](#))

4. Center for Cellular and Molecular Biology, Hyderabad, India Master Thesis, 2005

Dissertation title: *Y chromosome and mitochondrial DNA (mtDNA) based phylogenetic analysis of Ho Tribe of Jharkhand [BMC Evolutionary Biology 2008, 8:227]*

MENTORSHIP EXPERIENCE

1. Served as a teaching assistant to Graduate students in Pathogen Genomics course in KAUST.
2. Training medical students/fellows on Next-generation sequencing bioinformatics.

3. Supervising Master students for research theses from different universities.

CONTRIBUTION TO SCIENCE

1. **Discovery of a novel potentially transforming somatic mutation in breast cancer tissue.**

We discovered recently a new somatic mutation potentially conferring ligand-independence to breast cancer tissue or cell line from a breast cancer patient. I have developed the analysis pipeline and led the computational part of the study. This work is accepted for publication in Cancer Medicine in June 2021. [https://doi.org/10.1002/cam4.4106]

2. **Identification of therapeutic targets in Heat Stroke patients (pilgrims from Makkah) using multi-omics dataset.**

I have been working on a unique project to decipher human heat stress response mechanism or pathways together with Dr. Abderrezak Bouchama (Experimental Medicine Department, KAIMRC) for last three years. My role on this project is to develop and conduct transcriptome, proteome, metabolome, and whole genome sequence analyses on data derived from heat stroke patients and heat stress individuals from the pilgrims appear in Makkah every year. We did identify some interesting therapeutic/prognostic genes/pathways involved in pathophysiology of this disease. We are embarking on patenting the novel findings from this project. [https://doi.org/10.1113/JP284031]

3. **Development of a clinical-grade sequencing pipeline for genetic variants discovery in patients with genetic disorders.**

Together with Dr. Majid Al Fadhel from Medical Genomics in KAIMRC we developed a pipeline for whole genome/exome sequencing genetic disorder patients for the diagnosis and treatment. I have been leading the sequencing analysis and thus developed a clinical-grade sequencing pipeline to accurately identify causal genetic variants in this disease model. The computational pipeline validation results are significant and the outcome is used for diagnosis of the genetic disorders. [https://doi.org/10.1186/s12920-020-00743-8]

4. **Development of a computer algorithm/software for identification of pathogenic sequences in human genome sequences.**

During my post-doctoral research in KAUST I and Mr. Raece Naeem (a software programmer) developed a small program to identify the non-human sequence traces in the heap of high-throughput human sequencing data sets. This pipeline was used to quality check the single-molecule sequencing data in FANTOM5 project (Riken Genomics Science Center). This tool is available for public use as READSCAN [PMID: 23193222].

5. **Infectious disease research**

During my PhD I developed an artificial intelligence-based prediction model to predict the subcellular location of mycobacterial proteins [BMC Bioinformatics 2007, 8:337]. This system has been used for annotating the mycobacterial genome annotation.

AWARDS AND ACADEMIC ACHIEVEMENTS

- SABIC Post-Doctoral award (from January 2011 – January 2012) on the project “Creation of a High-throughput SNP Discovery Platform for Indian White Shrimp (*Fenneropenaeus indicus*)”. [KAUST, KSA]
- Recipient of “Council of Scientific and Industrial Research” (CSIR) Junior and Senior Research Fellowship (JRF & SRF) for the period of 5 yrs during PhD (2005-2010). [India]
- Scored 92.3 percentile in “Graduate Aptitude Test in Engineering” (GATE) 2005. [India]

PUBLICATIONS

1. **Rashid M**, Saha S, Raghava GPS: Support Vector Machine-based method for predicting subcellular localization of mycobacterial proteins using evolutionary information and motifs. **BMC Bioinformatics** 2007, 8:337.
2. G. Chaubey, M. Karmin, E. Metspalu, M. Metspalu, D. Selvi-Rani, V. Singh, J. Parik, A. Solnik, B. P. Naidu, A. Kumar, N. Adarsh, C. Mallick, B. Trivedi, S. Prakash, R. Reddy, P. Shukla, S. Bhagat, S. Verma, S. Vasnik, I. Khan, A. Barwa, D. Sahoo, A. Sharma, **M. Rashid**, V. Chandra, A. Reddy, A. Torroni, R. Foley, K. Thangaraj, L. Singh, T. Kivisild and R. Villems: Phylogeography of mtDNA haplogroup R7 in the Indian peninsula. **BMC Evolutionary Biology** 2008, 8:227
3. **Rashid M**, Singla D, Sharma A, Kumar M, Raghava GPS: Hmrbase: a database of hormones and their receptors. **BMC Genomics** 2009, 10(1): 307.
4. **Rashid, M.**, S. Ramasamy, et al. (2010). "A Simple Approach for Predicting Protein-Protein Interactions." **Current Protein & Peptide Science** 11(7): 589-600.
5. Abdallah M. Abdallah, **Mamoon Rashid**, Sabir A. Adroub, Marc Arnoux, Shahjahan Ali, Dick van Soolingen, Wilbert Bitter and Arnab Pain (2012). Complete Genome Sequence of Mycobacterium phlei type strain-RIVM1010601174. **Journal of Bacteriology** 194(12): 3284-3285.
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7. Naeem, R., **M. Rashid**, et al. (2012). "READSCAN: A fast and scalable pathogen discovery program with accurate genome relative abundance estimation." **Bioinformatics**. doi: 10.1093/bioinformatics/bts684
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10. Francy Jimenez-Infante, David Kamanda Ngugi, Intikhab Alam, **Mamoon Rashid**, Wail Baalawi, Allan Anthony Kamau, Vladimir B. Bajic, and Ulrich Stingl (2014). Genomic differentiation among two strains of the PS1 clade isolated from

- geographically separated marine habitats. *FEMS Microbial Ecology* doi: 10.1111/1574-6941.12348.
11. David Kamanda Ngugi, Jochen Blom, Intikhab Alam, **Mamoon Rashid**, Wail Ba-Alawi, Guishan Zhang, Tyas Hikmawan, Yue Guan, Andre Antunes, Rania Siam, Hamza El Dorry, Vladimir Bajic, and Ulrich Stingl (2014). "Comparative genomics reveals adaptations of a halotolerant thaumarchaeon in the interfaces of brine pools in the Red Sea." *ISME J* 9(2): 396-411.
 12. Sharma, A., D. Singla, **M. Rashid** and G. P. Raghava (2014). "Designing of peptides with desired half-life in intestine-like environment." *BMC Bioinformatics* 15(1): 282.
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 15. **Mamoon Rashid** and Ulrich Stingl (2015). "Contemporary molecular tools in microbial ecology and their application to advancing biotechnology" *Biotechnol Adv* 33(8): 1755-1773.
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POSTERS AND ORAL PRESENTATIONS

- Conducted workshop on "AI in Genomics" in Health Innovation Hackathon, King Saud bin Abdulaziz University for Health Sciences, Riyadh, 2023
- **eQTL: Finding regulatory SNPs in Heat Stroke.** 01/12/2021, King Abdullah International Medical Research Center, National Guard, Riyadh
- **Bioinformatics method for identification of somatic mutations in cancer and their functional relevance.** 02/25/2020, 2nd Therapeutic Discovery Conference, King Abdullah International Medical Research Center, National Guard, Riyadh
- **Establishment of Whole Exome and Whole Genome sequencing workflow for variant discovery in genetic disorder patients.** 02/13/2018, King Abdullah Specialist Children Hospital, National Guard, Riyadh

- **Computational discovery/identification of pathogens from sequence data.** 2/21/2017, King Abdullah International Medical Research Center, National Guard, Riyadh
- Poster presentation for King Abdullah University of Science and Technology (KAUST) president's International Advisory Committee in February 2011, Thuwal on "**Creation of a High-throughput SNP Discovery Platform for Indian White Shrimp (*Fenneropenaeus indicus*)**".
- Presentation in FANTOM5 consortia in RIKEN Institute, Yokohama, Japan on "**Using epigenetic marks to improve detection of tissue specific hCAGE-tag clusters**" on October 2011.
- Presentation in FANTOM5 consortia in RIKEN Institute, Yokohama, Japan on "**Tissue-specific expression of transcription factors and co-factors in FANTOM5 dataset**" on January 2011.
- Delivered several lectures and tutorials as a part of **Open Source for Drug Discovery** (a Government of India initiative) in Institute of Microbial Technology, Chandigarh, India, 2008.

REFERENCES

References will be provided upon request.